

Stage 06 — Exploratory Data Analysis (EDA)

Telecom Customer Churn Analysis

This stage moves from data cleaning into exploration: understanding customer behavior, churn patterns, and variable relationships in order to generate stronger questions for the diagnostic-analysis phase that follows.

EDA identifies patterns and associations; it does not prove causation.

Goal & Scope

Explore the cleaned dataset (produced in Stage 05), understand customer distributions and variable behavior, identify visible patterns and relationships, and generate hypotheses for the later business-question and diagnostic stages. This notebook uses the cleaned dataset — it does not modify the source data or make final business recommendations. Final answers to predefined business questions remain in Stage 07.

Recommended EDA Structure

Step	What Needs to Be Executed	Tools
6.1 Load Clean Dataset	Load the validated dataset produced in Stage 5.	Python, Pandas
6.2 Dataset Overview	Review shape, columns, data types, and summary statistics.	Pandas
6.3 Customer Base Profile	Explore the overall customer population and major categorical distributions.	Pandas, Matplotlib, Seaborn
6.4 Target Distribution	Explore overall churn counts and percentages.	Pandas, Matplotlib/Seaborn
6.5 Numerical Distributions	Explore tenure, monthly_charges, and total_charges.	Pandas, Matplotlib, Seaborn
6.6 Categorical Distributions	Visualize major customer/service/contract/payment categories.	Matplotlib, Seaborn
6.7 Churn vs Numerical Variables	Explore how churned and retained customers differ in tenure and charges.	Pandas, Matplotlib, Seaborn
6.8 Churn vs Categorical Variables	Explore visible churn differences across major customer dimensions.	Pandas, Matplotlib, Seaborn
6.9 Relationship Analysis	Explore relationships among important numerical variables.	Pandas, Seaborn
6.10 Correlation Analysis	Examine numerical correlations where meaningful.	Pandas, Seaborn
6.11 Distribution/Pattern Review	Identify unusual patterns that deserve investigation later.	Pandas, Matplotlib, Seaborn
6.12 EDA Findings Log	Document observations and potential questions without turning them into conclusions.	Markdown/Jupyter

Notebook Workflow (as implemented)

1. Import Libraries

Loads pandas, numpy, matplotlib, and seaborn; sets a consistent Seaborn theme and displays all DataFrame columns.

2. Load the Cleaned Dataset

Reads the Stage 05 cleaned CSV so EDA does not repeat the cleaning process.

3. Dataset Overview

Baseline population statistics: customer count, feature count, and average tenure, monthly charges, and total charges — the reference point for later segment comparisons.

4. Churn Overview

Overall churn counts and percentages, visualized with a count plot, to establish the class imbalance in the customer population.

5. Churn Rate vs. Churn Count

Uses a normalized crosstab and a groupby-mean on contract type to express churn as a rate within each group rather than a raw count, which is more meaningful for risk comparison.

6. Tenure EDA

Summary statistics, a histogram of tenure, and a boxplot comparing tenure between churned and retained customers. Shorter tenure among churners would point to early lifecycle stages as worth deeper diagnostic analysis.

7. Monthly Charges EDA

Summary statistics plus a histogram, boxplot, and KDE plot comparing monthly charge distributions by churn status.

8. Total Charges

Boxplot comparing total charges by churn status, with a caution that total charges are strongly tied to tenure since charges accumulate over time — differences shouldn't be read as a churn cause.

9. Contract EDA

Customer counts/percentages and churn rate by contract type, with a count plot for distribution.

10. Payment Method EDA

Churn rate by payment method, visualized as a horizontal bar chart sorted by rate.

11. Internet Service EDA

Churn rate by internet service type, visualized with a bar plot.

12. Service-Level Analysis

A loop over six optional service columns (online security, online backup, device protection, tech support, streaming TV, streaming movies) applying the same churn-rate calculation consistently rather than duplicating code per service.

13. Demographic EDA

Churn rate by senior-citizen status, partner status, and dependent status — useful for profiling, but a difference alone doesn't make a variable a strong or actionable driver.

14. Numeric Comparison by Churn

Mean and median tenure, monthly charges, and total charges compared between churned and retained groups.

15. Correlation Analysis

Correlation matrix across senior_citizen, tenure, monthly_charges, total_charges, and churn_flag, visualized as a heatmap. Correlation measures association, not causation, and total charges/tenure are naturally related.

16. Questions Generated by EDA

A list of candidate questions carried forward into the next phase (see below), rather than conclusions drawn prematurely.

Key Interpretation Cautions

- EDA identifies patterns and associations — it does not prove causation.
- Total charges are strongly related to tenure because charges accumulate over time; a difference in total charges should not automatically be read as a churn cause.
- Demographic differences (e.g., senior citizen, partner, dependents) help profile customers but don't automatically make a variable a strong or actionable churn driver.
- Correlation measures association, not causation.

Questions Generated for the Next Phase

- 1 Is early-tenure churn concentrated among month-to-month customers?
- 2 Are high monthly charges associated with churn within the same contract type?
- 3 Is a high-churn payment method explained by its contract mix?
- 4 Does the number of subscribed support/security services relate to churn?
- 5 Which combination of tenure group, contract, and monthly-charge band has the highest churn rate?
- 6 Which high-churn segments are also large enough to represent a meaningful retention opportunity?

Next stage: Stage 07 will use these candidate questions to drive targeted, diagnostic analysis and answer the project's predefined business questions.